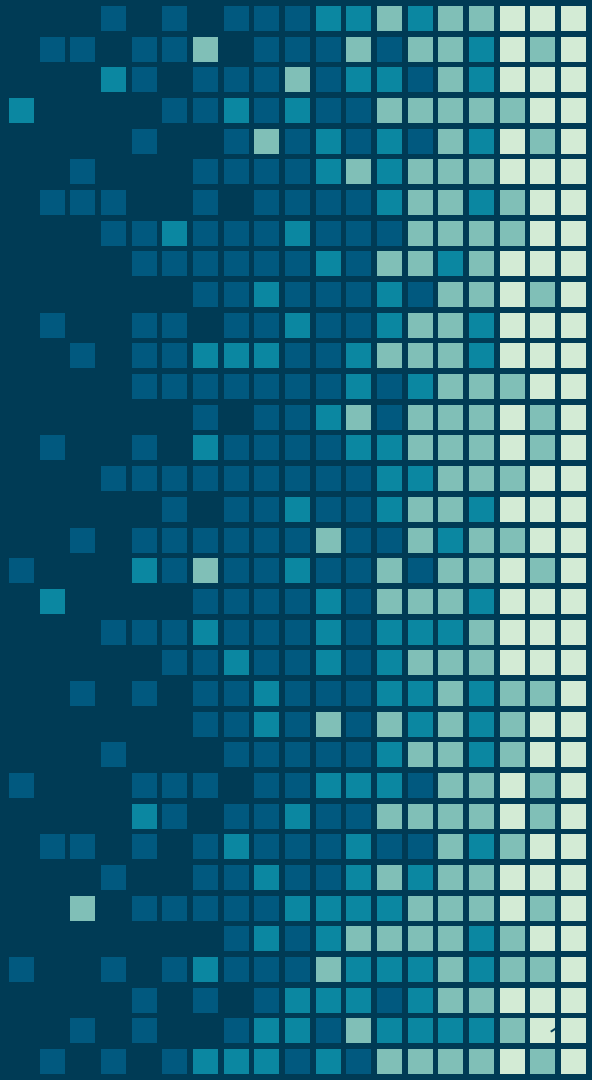
EMBEDDED CONTROL

Prof. Luca De Cicco
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https://c3lab.poliba.it/LDC

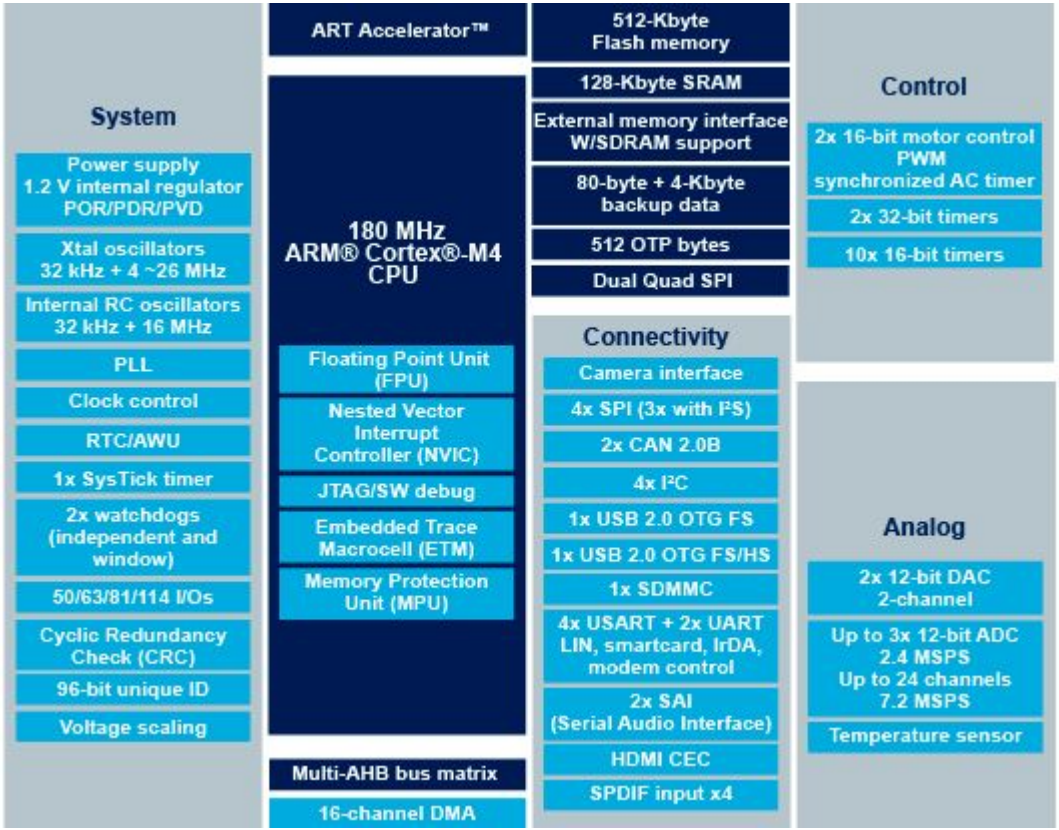


STM32F446RE Nucleo-64 dev boards

The STM32F446RE microcontroller

- Based on ARM Cortex-M4 32-bit RISC core operating at a frequency of up to **180 MHz**
- The Cortex-M4 core features a Floating point unit (FPU) single precision
- Implements a full set of Digital Signal Processing instructions
- Implements memory protection unit (MPU)
- 512 KB of flash memory (code)
- 128 KB of SRAM (volatile data)
- 64 pins package LQFP64 (1cm x 1cm)

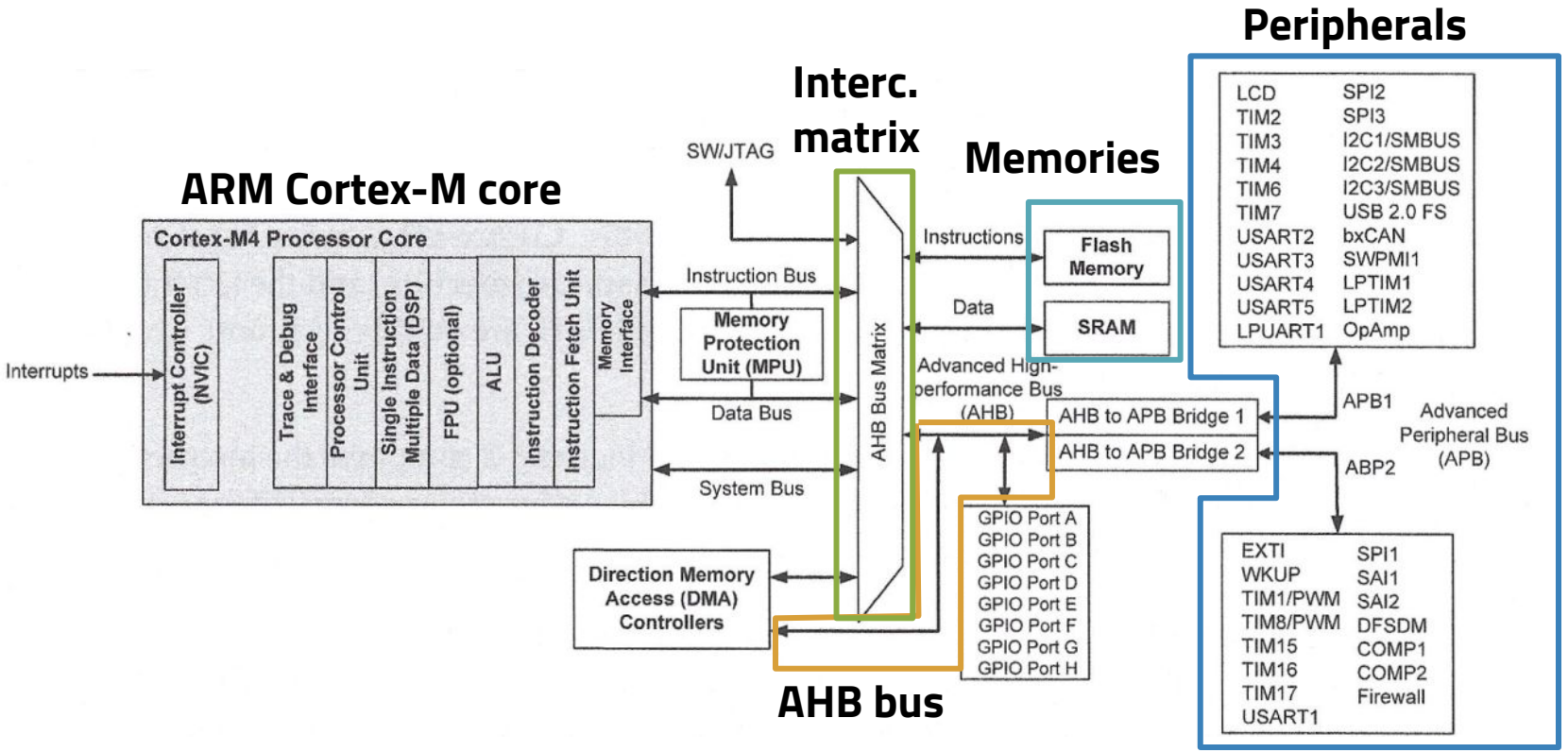
STM32F446RE features



STM32F446RE components

- The **ARM Cortex-M4** processor connected to an interconnection bus matrix (more on that) through which it is connected to:
- AHB (high perf) and APB1/APB2 buses (for periphs)
- A bridge connecting the AHB to the periph buses APB1/APB2
- Memories (flash, SRAM, external memories)
- **Direct Memory Access** (DMA) controller to support data transfers between memories and peripherals
- **General-Purpose Input Output** (GPIO) ports to connect the MCU to external devices and are physically mapped to an MCU pin!
- A GPIO have multiple configurable functions (via registers)
- **Peripherals**: ADC, DAC, Timers, watchdog, I2C, USART, SPI, CAN
- Each peripheral is accessible through memory mapped registers

Block diagram STM32F4 (simplified)



System architecture

32-bit multilayer AHB bus matrix interconnecting:

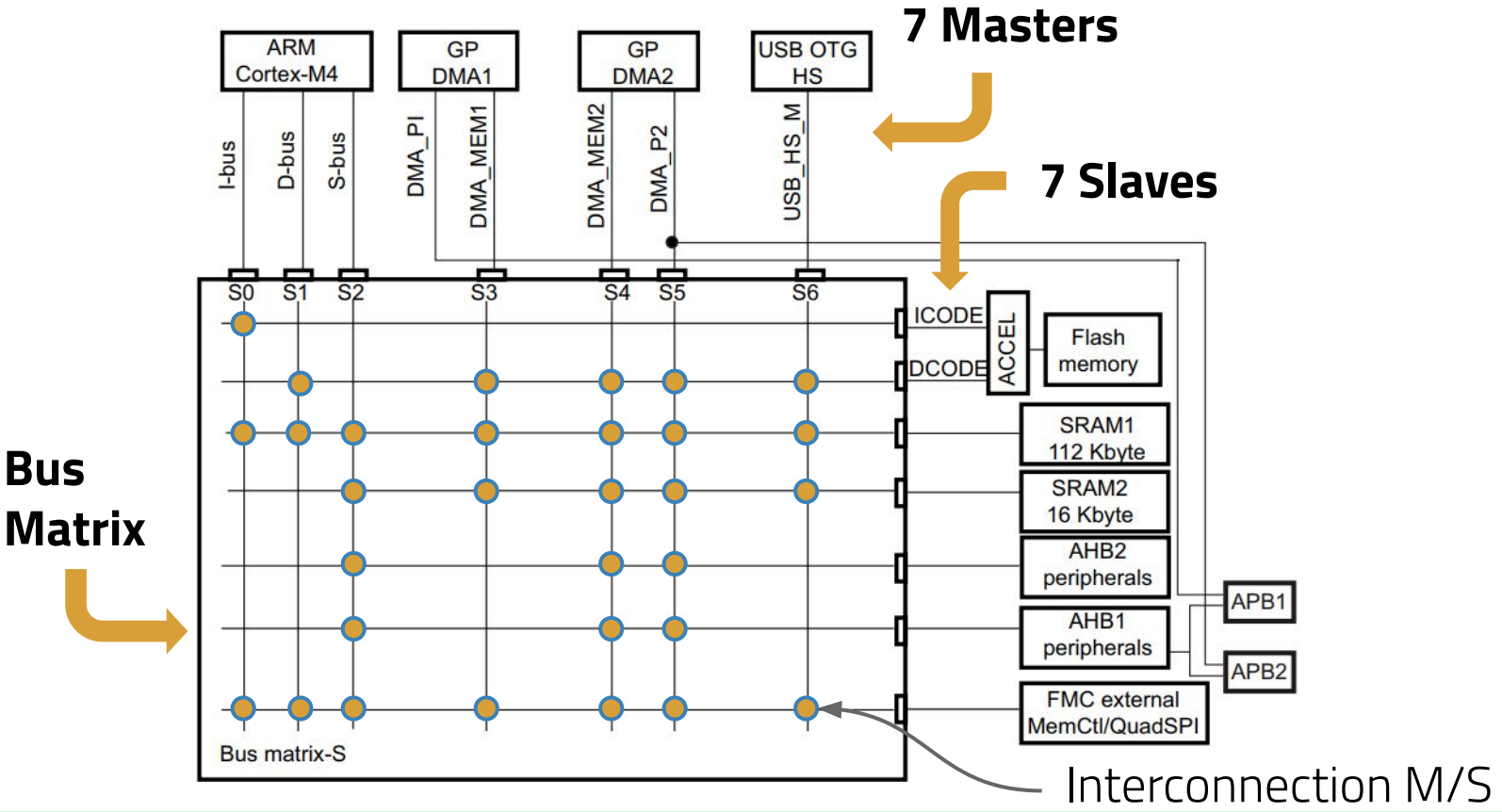
7 masters buses:

- Three Cortex-M4 buses: **I-Bus**, **D-Bus**, **S-Bus** (sys bus)
- **DMA1**, **DMA2** memory buses
- **DMA2** peripheral bus
- **USB** OTG HS DMA bus

7 slaves:

- Internal Flash memory **ICode**/**DCode** bus
- Main internal **SRAM1**, auxiliary **SRAM2**
- **AHB1** (includes the APB bus through bridge)**, AHB2** peripherals
- **QUADSPI** (used to attach external memories)

AHB Bus Matrix

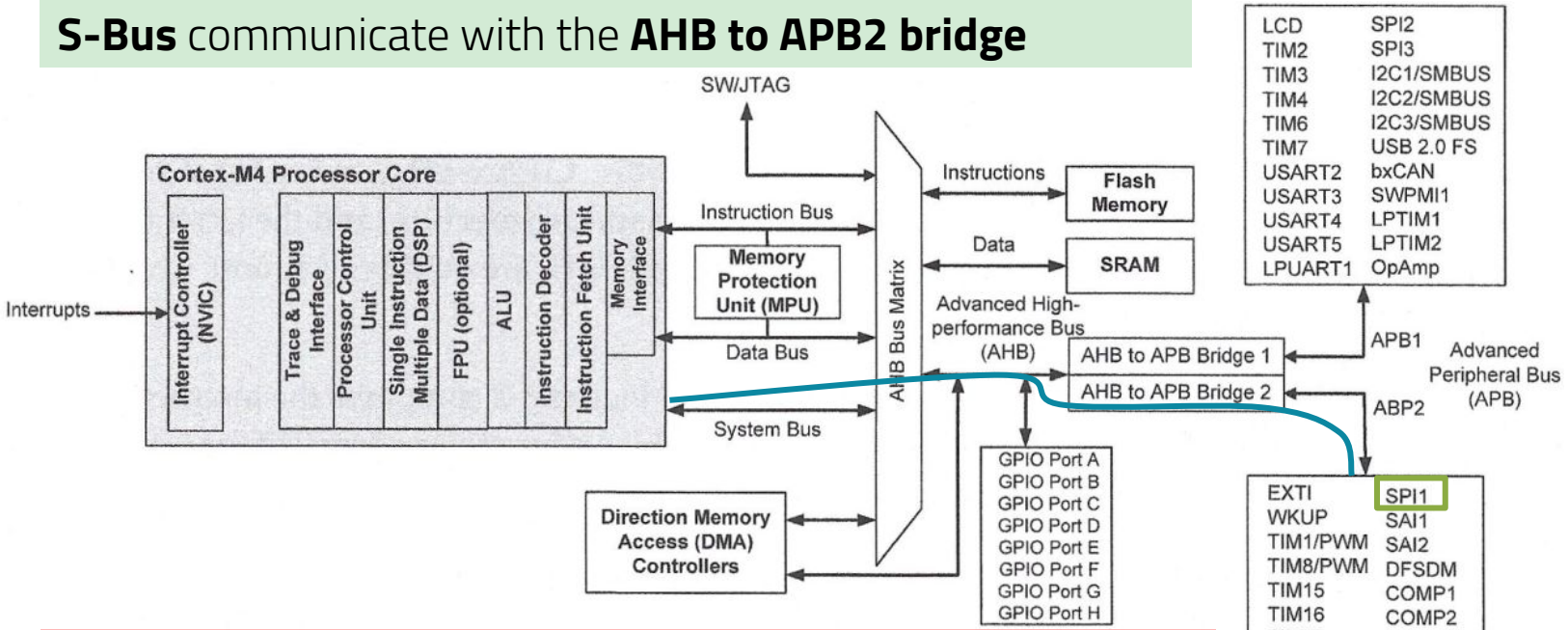


Main Buses

- **I-bus**: used by the core to fetch instructions. The target of I-bus is a memory containing code (could be internal Flash memory/SRAM or external memories through the QUADSPI iface)
- **D-bus**: used by the core for data load and debug. The target of this bus is a memory containing code or data (same of I-Bus)
- **S-Bus** (system bus): used to access data located in a peripheral or in SRAM. Targets: internal SRAM, SRAM2, the AHB1 peripherals including the APB peripherals, the AHB2 peripherals and the external memories through the FMC and QUADSPI
- **DMA memory/periph**: used by the DMA to perform transfer to/from memories and periphs. Targets: internal Flash, internal SRAMs and external memories; all peripherals (AHB, APB)

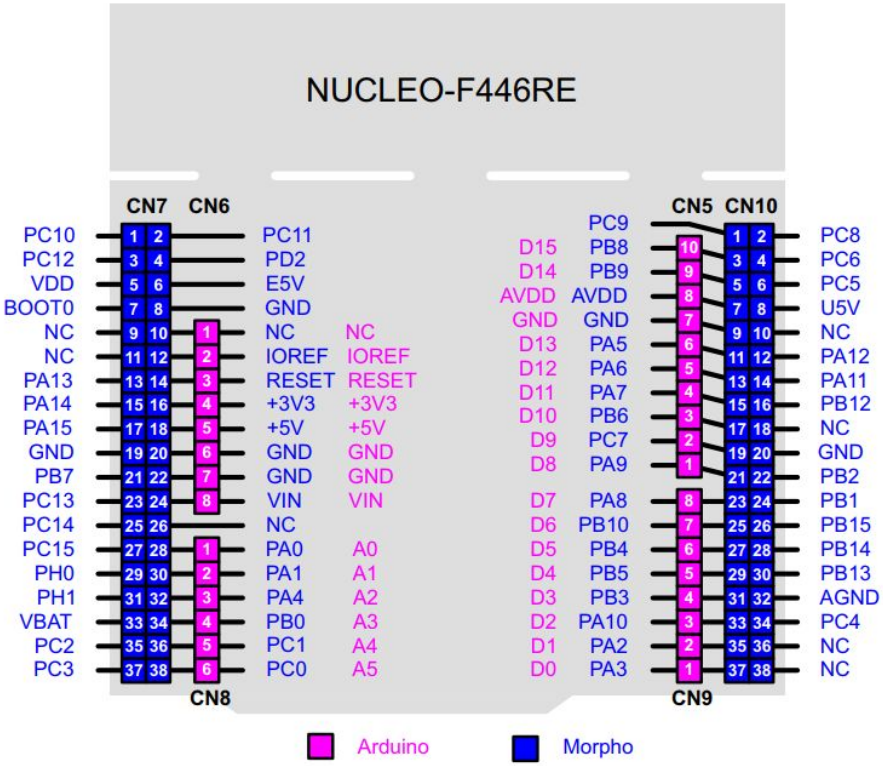
Example: Core access data from **SPI1** peripheral

At HW level: Interconnection matrix is used to make the **S-Bus** communicate with the **AHB to APB2 bridge**

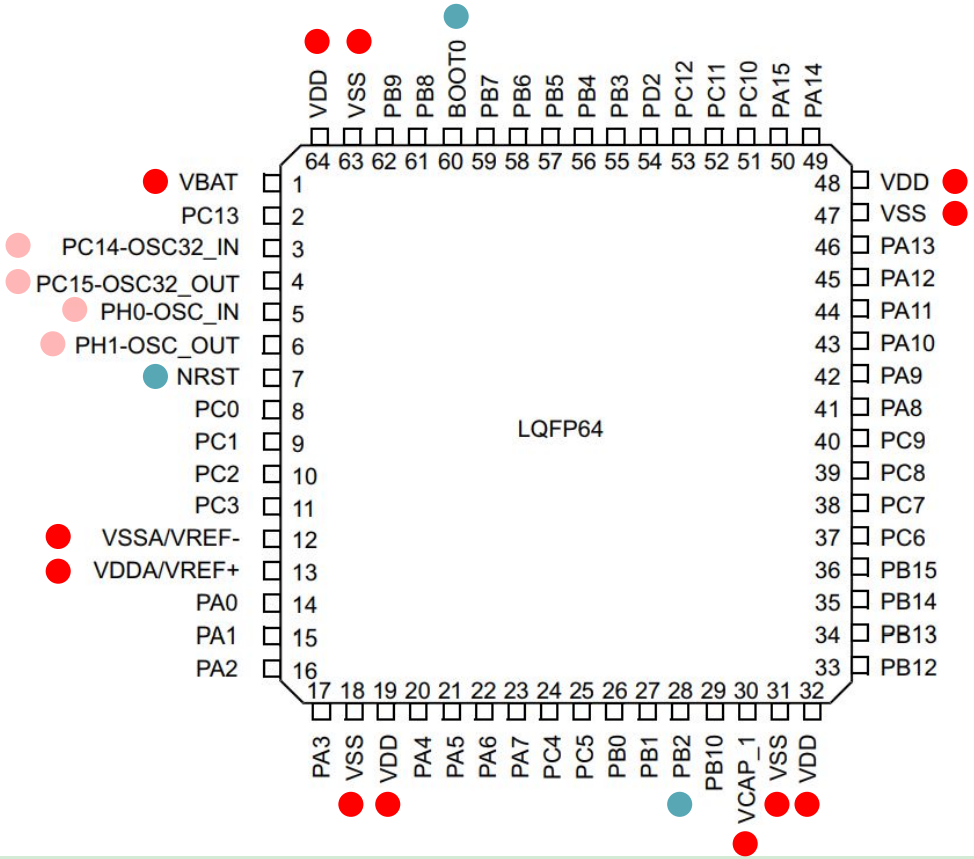


Transparent from the programmer point of view. To access SPI1 you only need the address of the registers!

MCU to external devices (for reference)



STM32F446RE (LQFP**64** 10x10mm package)

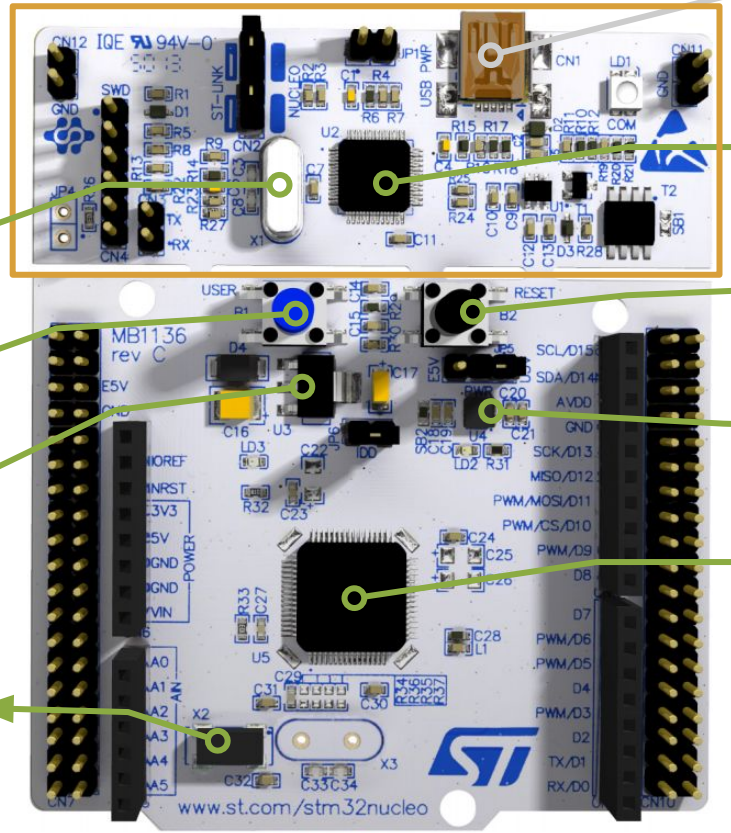


- Power supply
- Pins to boot (BOOT0, PB2 AF) or reset (NRST)
- Pins to connect external oscillators
- All the other pins Pxy are dedicated to GPIO or their alternate functions

MCU integration

- An MCU alone serves no purposes
- You need power supply (voltage regulators)
- Connectors to connect to the external devices
- Crystals (optional but recommended) for clock generation
- Buttons to serve specific purposes (f.i., to reset the MCU)
- Any other electronic devices needed for your application

NUCLEO64 Boards anatomy



mini USB
(Power, programming)

ST-LINK
Programmer

STLINK MCU

Ext. oscillator
(8 MHz)

Reset Button (B2)

User Button (B1)

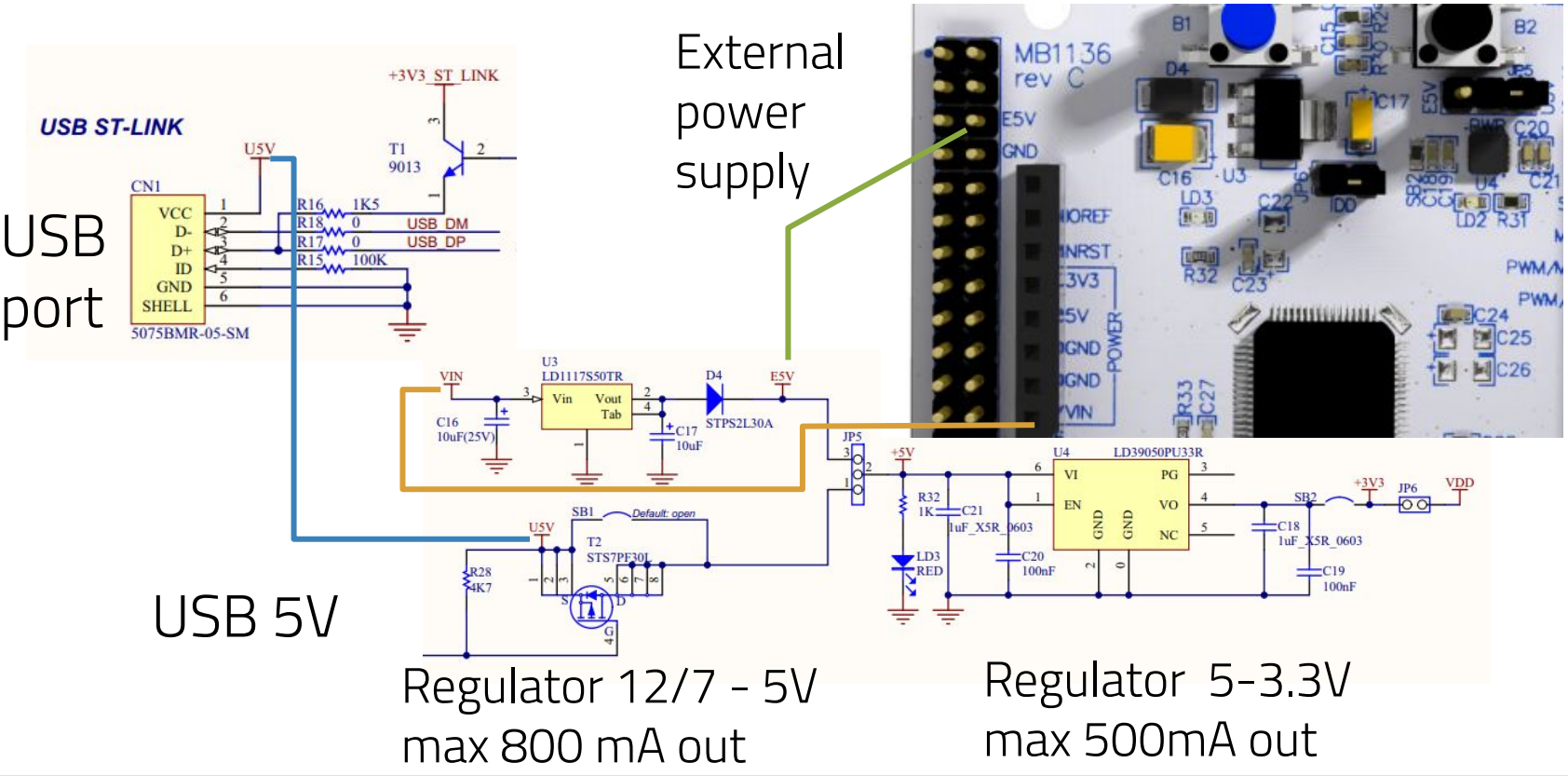
Voltage Regulator
(~5VIN - 3.3 VOUT)

Voltage Regulator
(12-7VIN - 5VOUT)

MCU
STM32F446RE

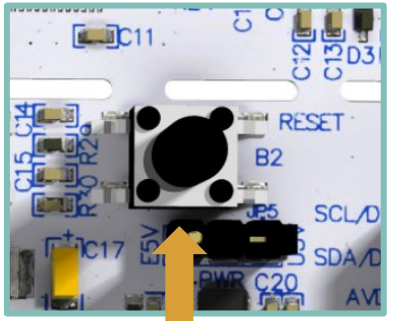
Ext. oscillator
(32 KHz)

Power supply schematics (UM1724)



Power supply connection to MCU

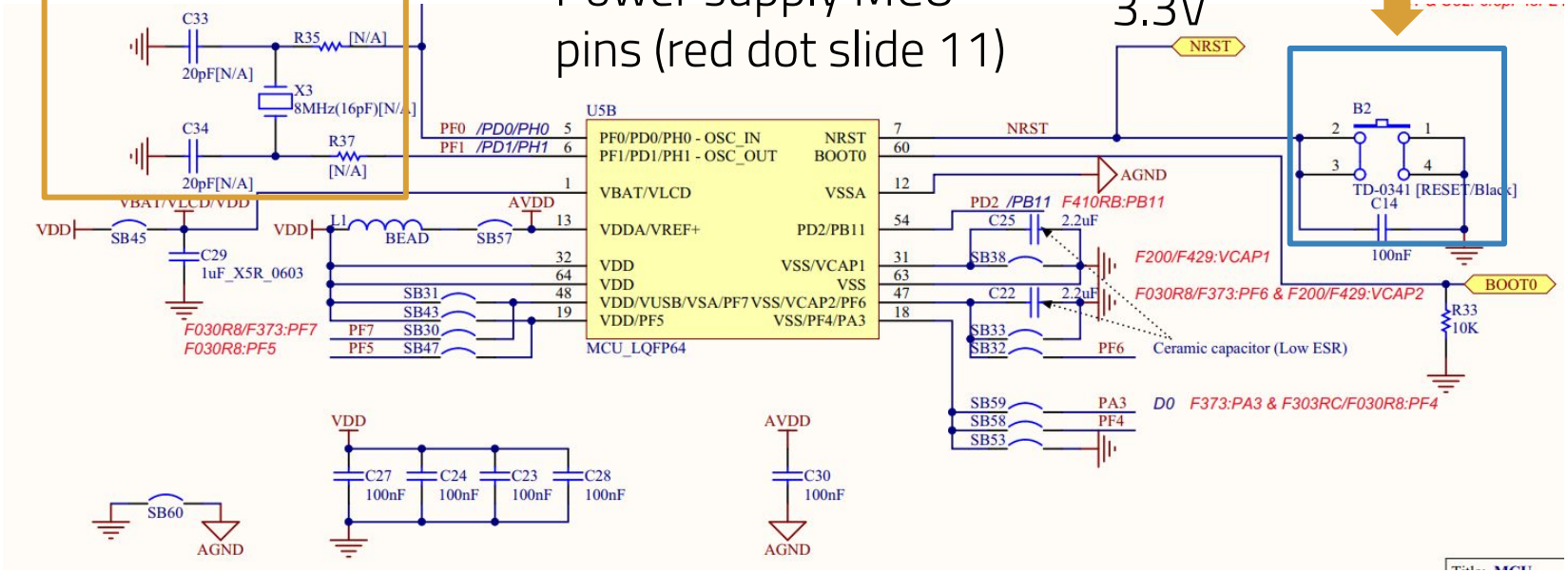
Reset button wired to NRST



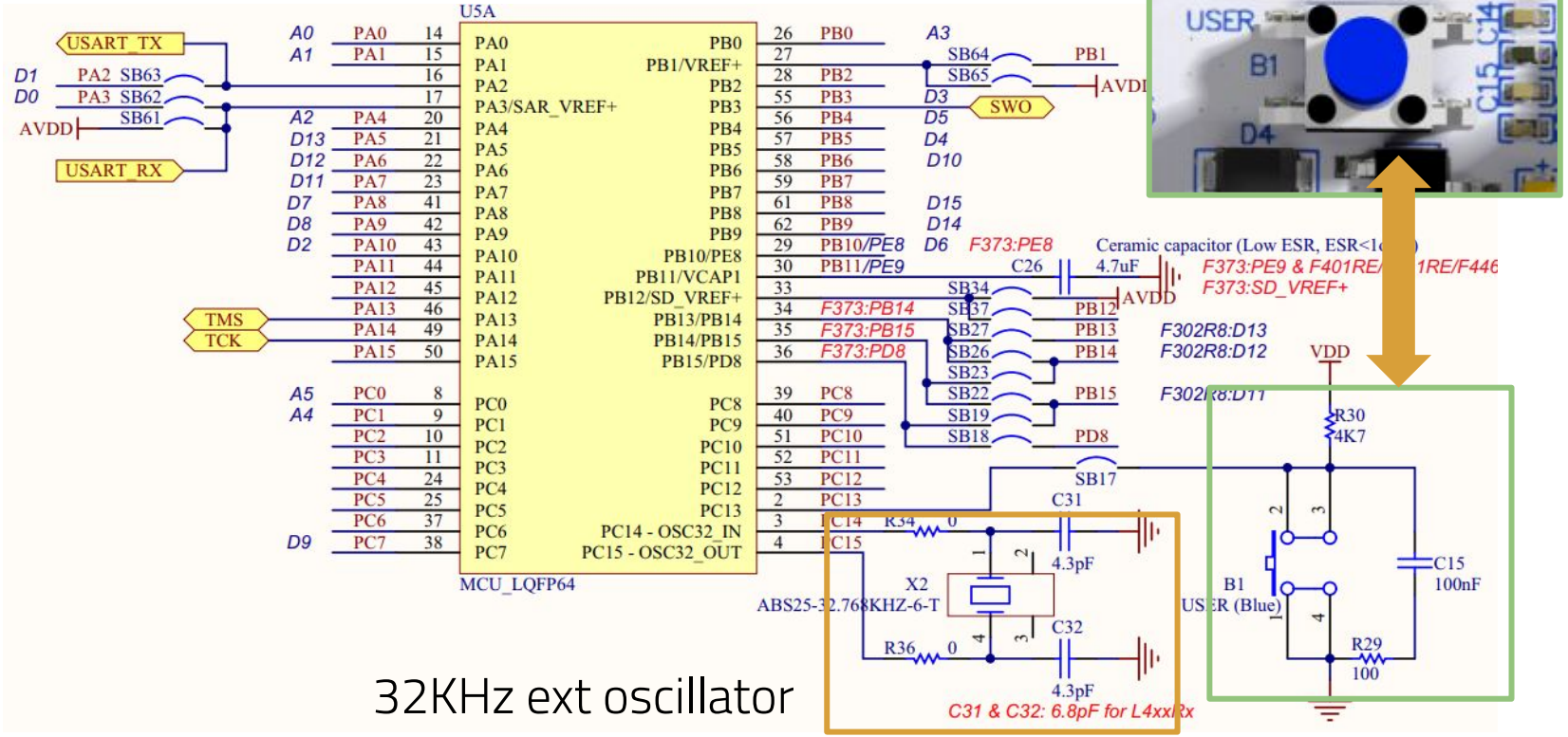
8 MHz ext oscillator

Power supply MCU pins (red dot slide 11)

Normally 3.3V



MCU to external devices



32KHz ext oscillator

References

- Nucleo64 boards user manual
- Schematics (check the PDF in the zip achieve)

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